

LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 3

Course Code: INT396

Course Title: UNSUPERVISED LEARNING

Course Outcome: CO3

Paper: Sample Multiple Choice Questions

Maximum Marks: 40 x 1 = 40

Unit: 3

Unit Title: Hierarchical and Density-Based Clustering

Unit Mapped RBT: L3 (Apply)

Total Questions: 40

Suggested Duration: 60 Minutes

Answer Grid (to be filled by the student)

Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
Q1		Q11		Q21		Q31	
Q2		Q12		Q22		Q32	
Q3		Q13		Q23		Q33	
Q4		Q14		Q24		Q34	
Q5		Q15		Q25		Q35	
Q6		Q16		Q26		Q36	
Q7		Q17		Q27		Q37	
Q8		Q18		Q28		Q38	
Q9		Q19		Q29		Q39	
Q10		Q20		Q30		Q40	

Multiple Choice Questions (40 x 1 = 40 Marks)

Q1. Name the hierarchical approach that begins with every record in one cluster and splits it repeatedly.

- A) Divisive clustering
- B) Agglomerative clustering
- C) Density-based clustering
- D) Partitional clustering

Q2. Identify the label a DBSCAN implementation conventionally gives to a noise point.

- A) 0
- B) -1
- C) 1
- D) Null

Q3. State the linkage criterion that merges the pair of clusters producing the smallest increase in the total within-cluster variance.

- A) Single linkage
- B) Complete linkage
- C) Ward linkage
- D) Average linkage

Q4. Name the table of pairwise distances that agglomerative clustering has to hold while it runs.

- A) The confusion matrix
- B) The covariance matrix

C) The transition matrix

D) The proximity matrix

Q5. Identify the plot conventionally used to choose a value of eps for DBSCAN.

- A) The scree plot
- B) The k-distance plot
- C) The elbow plot of inertia
- D) The silhouette plot

Q6. State the number of merges agglomerative clustering performs on n records before a single cluster remains.

- A) n
- B) n + 1
- C) n / 2
- D) n - 1

Q7. Explain why agglomerative clustering does not require the number of clusters before it runs.

- A) It builds the whole hierarchy and is cut afterwards
- B) It discovers the number from the density of points
- C) It always reports exactly two final clusters
- D) It requires the number just as k-Means does

Q8. Distinguish agglomerative clustering from divisive clustering by the direction in which each one builds.

- A) Divisive builds upwards, agglomerative builds downwards
- B) Both of the two build upwards from single records
- C) Agglomerative builds upwards, divisive builds downwards
- D) Both of the two build downwards from one cluster

Q9. Describe what a horizontal cut placed across a dendrogram produces.

- A) A single cluster containing every record
- B) A ranking of the records by their height
- C) A partition whose size is the number of lines crossed
- D) A list of the records labelled as noise

Q10. Interpret a dendrogram on which nearly every merge takes place at a similar height.

- A) The data shows little clear group structure
- B) The data holds exactly two natural groups
- C) The linkage criterion was computed wrongly
- D) The records are duplicates of one another

Q11. Predict the effect on a DBSCAN result of raising MinPts while eps is held fixed.

- A) More core points and less noise
- B) The same clusters with new labels
- C) One single cluster containing everything
- D) Fewer core points and more noise

Q12. Recognize the linkage criterion that measures the distance between two clusters as the distance between their two furthest members.

- A) Single linkage
- B) Complete linkage
- C) Average linkage
- D) Ward linkage

Q13. Explain why single linkage produces the effect known as chaining.

- A) It requires every pair of points to be close
- B) It merges the two largest clusters at each step
- C) It measures distance from the cluster centres
- D) One close pair is enough to join two groups

Q14. Describe what the eps-neighbourhood of a point contains.

- A) Every point in the same cluster as it
- B) The MinPts points nearest to it
- C) Every point lying within distance eps of it
- D) Every point that is not labelled noise

Q15. Explain why the cluster a DBSCAN border point ends up in can depend on the order the records are processed.

- A) A border point is assigned entirely at random
- B) A border point may lie near two different core points
- C) A border point is always relabelled as noise
- D) A border point changes the value of eps

Q16. Distinguish the memory agglomerative clustering needs from the memory k-Means needs.

- A) Agglomerative grows with the square of the records
- B) k-Means grows with the square of the records
- C) Both of the two grow with the square of the records
- D) Neither of the two depends on the record count

Q17. Interpret a DBSCAN result that consists of one very large cluster with no noise points at all.

- A) The value of eps has been set too small
- B) The value of eps has been set too large
- C) The value of MinPts has been set too large
- D) The dataset contains no genuine structure

Q18. Predict where two identical records will be joined on a dendrogram.

- A) At a height of zero, before any other merge
- B) At the very top, after every other merge
- C) At the average height of the other merges
- D) They are never joined to one another

Q19. Solve the following: agglomerative clustering begins with 30 separate records. Determine how many clusters remain after 25 merges.

- A) 6
- B) 25
- C) 29
- D) 5

Q20. Use single linkage to determine the distance between two clusters whose four inter-cluster pairwise distances are 8, 3, 6 and 5.

- A) 8
- B) 5.5
- C) 3
- D) 6

Q21. Apply average linkage to two clusters holding 3 and 4 records whose twelve pairwise distances sum to 60.

- A) 5
- B) 12

- C) 7.5
- D) 20

Q22. Compute the number of distinct pairwise distances that must be held for 200 records.

- A) 40,000
- B) 20,000
- C) 19,900
- D) 200

Q23. Apply the DBSCAN definition with MinPts set to 5 to a point that has 6 points inside its eps-neighbourhood, itself included, and determine its status.

- A) A border point
- B) A core point
- C) A noise point
- D) It cannot be decided

Q24. Examine a point that has only 2 neighbours inside its eps-neighbourhood with MinPts set to 5, but which lies inside the eps-neighbourhood of a core point, and determine its status.

- A) A core point
- B) A noise point
- C) A centroid
- D) A border point

Q25. Use a dendrogram cut that crosses four vertical lines to determine the number of clusters reported.

- A) 2
- B) 5
- C) 4
- D) 8

Q26. Compute the number of merges still to be performed when agglomerative clustering of 50 records has reached 12 clusters.

- A) 12
- B) 38
- C) 49
- D) 11

Q27. Apply complete linkage to two clusters whose four inter-cluster pairwise distances are 2, 9, 4 and 7.

- A) 9
- B) 2
- C) 5.5
- D) 4

Q28. Compare single linkage with Ward linkage on the shape of cluster each one tends to produce.

- A) Single gives compact groups, Ward gives long chains
- B) Single gives long chains, Ward gives compact groups
- C) Both of the two give long straggling chains
- D) Both of the two give compact rounded groups

Q29. Use the k-distance plot to determine where the value of eps should be read.

- A) At the very start of the sorted curve
- B) At the very end of the sorted curve
- C) At the midpoint of the record count
- D) At the knee, where the distance begins to rise sharply

Q30. Examine a DBSCAN run in which eps has been set to a very small value, and determine the result.

- A) Every record falls into one cluster
- B) Almost every record is labelled as noise
- C) The clusters are unusually well separated
- D) The algorithm reports exactly two clusters

Q31. Apply DBSCAN to a dataset whose two groups form concentric rings and determine whether it can separate them.

- A) No, because it assumes convex clusters
- B) No, because it needs the number of clusters
- C) Yes, because it follows connected dense regions
- D) Yes, but only after the data is standardised

Q32. Compare the number of parameters that k-Means requires with the number that DBSCAN requires.

- A) k-Means needs one, DBSCAN needs two
- B) k-Means needs two, DBSCAN needs one
- C) Both of the two need exactly one
- D) Both of the two need exactly three

Q33. Compute the number of pairwise distances between a cluster holding 4 records and a cluster holding 5 records.

- A) 20
- B) 9
- C) 10
- D) 45

Q34. Use the comparison of the three clustering families to select a method for a dataset of irregular shapes and many stray records where the number of groups is unknown.

- A) k-Means
- B) k-Medoids

- C) MiniBatch k-Means
- D) DBSCAN

Q35. Examine the following and select the correct explanation: agglomerative clustering is run on 500,000 records and the machine runs out of memory before any result appears.

- A) The dendrogram cannot be drawn for so many records
- B) The proximity matrix grows with the square of the records
- C) The linkage criterion was chosen incorrectly
- D) The records were not standardised beforehand

Q36. Apply Ward linkage and determine what it minimises at each merge.

- A) The distance between the two nearest members
- B) The distance between the two furthest members
- C) The increase in the within-cluster sum of squares
- D) The number of records in the merged cluster

Q37. Demonstrate how the number of clusters is obtained from a completed dendrogram.

- A) Count the number of records at the bottom
- B) Count the total number of merges performed
- C) Place a horizontal cut and count the lines it crosses

- D) Read the height of the very last merge

Q38. Relate the value of eps to the number of clusters DBSCAN reports.

- A) A larger eps splits clusters into more of them
- B) A larger eps merges separate clusters into fewer
- C) The value of eps has no effect on the count
- D) The count is fixed before the algorithm runs

Q39. Use the definition of a core point to determine the status of a point that has one fewer neighbour than MinPts within eps and lies near no core point at all.

- A) A core point
- B) A border point
- C) A cluster centre
- D) A noise point

Q40. Compare hierarchical clustering with k-Means on whether repeated runs give the same answer.

- A) Hierarchical is deterministic, k-Means depends on its start
- B) k-Means is deterministic, hierarchical depends on its start
- C) Both of the two give a different answer each run
- D) Both of the two are completely deterministic

----- End of Sample Question Set -----